

Deep Learning-Enabled Intelligent Health Care System for Early Diagnosis of Alzheimer's Disease

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Abstract

Alzheimer's disease (AD) is the predominant form of cognitive decline, which slowly impacts on memory, thinking and behaviour over time. Timely recognition of AD is essential for prompt medical care and patient treatment. Current research outlines a comparative deep learning based healthcare system for automatic categorization of Alzheimer's disease stages using 3D T1-weighted Magnetic Resonance Imaging (MRI) scans from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. This research compares the efficacy of four distinct models - 3D DenseNet model with self-attention, XGBoost classifier, 3D ResNet18 model, and a radiomics-based XGBoost model. Overall, 403 MRI scans from three classes, Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer's disease (AD) were used. The pretrained 3D ResNet18 model, which outperformed the other models implemented, had an accuracy of 0.61, recall of 0.61, precision of 0.60, and F1-score of 0.60. The results show that transfer learning and 3D deep learning architectures are effective in learning structural brain abnormalities for the prediction of Alzheimer's disease progression.

Keywords: Alzheimer's Disease, Deep Learning, MRI, Transfer Learning, 3D CNN.

1 Introduction

Alzheimer's Disease (AD) is a growing neurological dysfunction that dramatically influence cognitive abilities and quality of life. It is a major neuro-cognitive impairment, and a health issue of substantial global significance, especially considering the swift aging of the individuals. Rapid and precise diagnosis is critical for efficient disease control and prompt clinical action. However, traditional diagnostic methods based on manual examination and interpretation of information from neuroimaging tests are often time-consuming, expensive, and require specialist expertise [Ali et al., 2024]. Emerging AI innovations, especially deep learning, have demonstrated their potential to automate healthcare imagery analysis. MRIs of the brain has been widely utilized to classify Alzheimer's disease in the past years, with the help of Convolutional Neural Networks (CNNs) that can retrieve relevant findings from the MRI and enhance diagnostic accuracy [LeCun et al., 2015, Litjens et al., 2017]. The vast majority of current methods are based on 2D slice-thinning processing, or region of interest (ROI) extraction, however, compromising the model's capacity to provide full structural information of the brain [Islam and Zhang, 2018, Marwa et al., 2023]. This constraint encourages methods that are able to make good use of the full volumetric data to better reflect the spatial dependencies related to disease progression. To overcome these limitations, this study utilizes 3D volumetric MRI data, instead of relying on a predefined set of slices and brain regions. It enables the model to identify both global and local structural patterns of the whole brain, and make a comprehensive view of the disease progression. Moreover, rather than using a single modeling strategy, this research conducts a careful comparative study of a variety of models under a common experimental framework.

The study evaluates and compare four different models - 3D DenseNet with self-attention, 3D ResNet18, XGBoost classifier, radiomics-based XGBoost model. These models reflect various aspects of the data, from statistical and texture-based features to deep hierarchical features, which have been handcrafted. This comparison aims to enhance comprehension of the effectiveness of various modelling approaches when tasked with multi-class classification of Alzheimer's disease stages as Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD). Our research findings show that the use of the pretrained 3D ResNet model successfully outperforms the other models with the best accuracy among the implemented models. The results show that transfer learning and volumetric feature learning are useful techniques in multi-class Alzheimer classification with MRI data. Overall, the proposed framework is a promising step towards creating more comprehensive, scalable, and clinically viable AI-driven diagnostic tools for Alzheimer's disease classification.

2 Background and Related Work

Enlarged ventricles, thinning of the cortex and the atrophy of the hippocampus are frequent abnormalities associated with the progression of Alzheimer's disease. These anatomical changes gradually affect cognitive functioning and are widely used as biomarkers for disease diagnosis and progression analysis [Cleveland Clinic, 2025]. Neuroimaging techniques play an important role in identifying these abnormalities and supporting clinical assessment of Alzheimer's disease stages [Medscape, 2022]. Magnetic Resonance Imaging (MRI) has emerged as the most common neuroimaging modality for Alzheimer's disease analysis because it offers deep structural insights about brain anatomy in a non-invasive manner [Ali et al., 2024]. MRI scans enable researchers and clinicians to observe anatomical changes associated with disease progression and differentiate among various disease stages. In this study, 3D T1-weighted MRI scans are utilized.

Several machine learning and deep learning techniques have been developed for Alzheimer's disease classification using MRI scans. Earlier studies mainly focused on traditional machine learning techniques that relied on handcrafted feature extraction from structural MRI scans. To identify Alzheimer's disease, [Davatzikos et al., 2008] suggested a pattern classification mechanism, utilizing high-dimensional MRI features and obtained classification accuracy of 80-85%. Likewise, the Support Vector Machine (SVM) based methods showed good performance in separating Alzheimer's patients from normal healthy individuals with an accuracy of about 85-90% in the experimental conditions used in the study [Klöppel et al., 2008]. Multiple machine learning algorithms were analyzed in another comparative study, with Generalized Linear Models (GLM) reaching an accuracy of 88.24% and k-Nearest Neighbors (kNN), Decision Trees, and Naïve Bayes reaching less than 86.41% [Shahbaz et al., 2019]. These approach yielded low classification accuracy, but were extremely sensitive to the quality and skill of handcrafted feature engineering and preprocessing.

An emerging area in research focuses on deep learning frameworks for automated feature retrieval and classification. The proposed multi-instance learning approach in [Liu et al., 2018] adopted a deep learning framework that extracted features from significant anatomical regions related to Alzheimer's disease and attained classification accuracy between 85–92%. Similarly, an ensemble-based deep learning framework utilizing densely connected CNN architectures achieved an accuracy of 93.18% for multi-stage Alzheimer's disease classification using MRI scans [Islam and Zhang, 2018].

A multi-branch CNN architecture proposed by [Dardouri, 2025] utilized extensive preprocessing, data augmentation, and stratified K-fold validation for MRI-based Alzheimer's disease classification and achieved a classification accuracy of 99.68%. However, the dataset used lacked sufficient real-world clinical diversity, limiting the generalization ability of the model. In another study, a 1D-CNN framework using smartphone accelerometer data for classification of Alzheimer's disease attained an accuracy of 90.91% and an F1-score of 0.897 [Jo et al., 2019]. The results provided promising findings, however the data size was small and the data was taken under controlled conditions.

Some research has investigated the multimodal learning methods, where MRI, PET and clinical information are used together with reasonably good diagnosis of Alzheimer's disease [Oxtoby and Alexander, 2017, Albright and Initiative, 2019]. These approaches offer complementary diagnostic information, but are costly for data acquisition and complicated by multimodal data alignment and preprocessing. Beyond that, missing data

in the different modalities and the need for high computations make them less scalable and less applicable in clinical practice.

3 Methodology

The proposed methodology is based on the comparison of four models: 3D DenseNet with a self-attention mechanism, a 3D ResNet18 model pretrained on the MONAI framework, a classifier based on XGBoost, and a radiomics based ML model. The workflow of each model is structured into four stages: Data Preprocessing, Feature Extraction, Model Training and Classification. All the experimentation and model training have been performed in the Kaggle cloud computing platform, equipped with NVIDIA Tesla T4 GPUs. It allowed the efficient training of computationally intensive 3D deep learning models on volumetric MRI data, including preprocessing, feature retrieval, and multi-class classification experiments. The computational environment not only made the training process quicker but also enabled the implementation of transfer learning and attention-based designs for Alzheimer's classification.

3.1 Dataset and Preprocessing

This research utilized data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database, a well-known benchmark databases in Alzheimer's disease analysis. The dataset is comprised of 3D T1-weighted MRI scans from multiple clinical sites and imaging protocols. Only T1-weighted MRI scans obtained using MPRAGE acquisition sequence and sagittal acquisition plane were chosen since they offer detailed anatomical information and a clear representation of brain tissues pertinent to the evolution of Alzheimer's Disease. This study involved three diagnostic categories: Cognitively Normal (CN), Mild Cognitive Impairment (MCI) and Alzheimer's Disease (AD). Following the filtering, 403 MRI scans remained (187 CN scans, 87 MCI scans, and 129 AD scans).

To ensure consistency in the representation of input information, all MRI scans were resized to fixed dimensions for model training. To eliminate the inter-scan variance and improve the numerical stability in the training process, intensity normalization was used. The image was filtered using Gaussian filter to reduce associated high frequency noise and retain important anatomical structures of the brain. The models were focused on relevant brain regions via skull stripping. Supervised data augmentation methods such as random rotation, flipping and affine transformations were also utilized to boost the model's generalization and to minimize overfitting during the training process. The min-max normalization was employed to normalize the intensities of MRI, as given in Equation. The dataset was partitioned into three subsets: training, validation, and testing sets, utilizing a stratified split ratio of 70:15:15 in order to keep same class size distribution in all three sets.

3.2 Implemented Models

3.2.1 3D DenseNet with Self-Attention Mechanism

The first model is based on the custom-designed 3D DenseNet model integrated with a self-attention mechanism for volumetric MRI classification. DenseNet has dense connections, meaning that feature maps from every previous layer are passed to every next layer, allowing efficient re-use of features and better gradient propagation. The network was trained directly on the full 3D MRI scans, enabling it to learn local and global patterns of the structure that are associated with Alzheimer's disease. To improve the model's capacity to detect the relevant spatial regions of the MRI volumes, a self-attention mechanism was added after the dense blocks. This mechanism computes attention weights that encapsulate long-range dependencies within the brain volume. The self-attention process is characterized as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

The extracted feature maps were aggregated using a Global Average Pooling layer succeeded by fully connected layers for multi-class classification. The model utilized the Adam optimizer, employed a categorical cross-entropy loss function, maintained a learning rate of 0.001, utilized a batch size of 8, and underwent 100 training epochs.

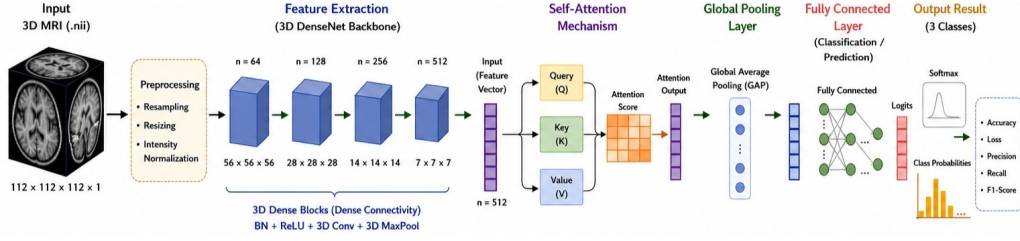


Figure 1: 3D DenseNet with Self Attention Architecture

3.2.2 XGBoost Classifier

The second implemented approach utilizes the Extreme Gradient Boosting (XGBoost) algorithm for Alzheimer's disease classification. Unlike deep learning models that learn features automatically from MRI volumes, the XGBoost classifier was trained on structured feature vectors derived from preprocessed MRI scans. Feature extraction and preprocessing operations were applied before classification to reduce dimensionality and improve classification efficiency. XGBoost optimizes a regularized objective function given by:

$$Obj = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (2)$$

where $l(\cdot)$ denotes the loss function and $\Omega(f_k)$ represents the regularization concept:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (3)$$

This formulation mitigates the probability of overfitting and control the model complexity. Each subsequent tree rectifies the error of its predecessors, using the technique of gradient boosting, and the model is trained iteratively. Optimal performance was obtained with the XGBoost classifier, trained with 400 estimators, learning rate of 0.03 and a maximum depth of 6 levels per tree.

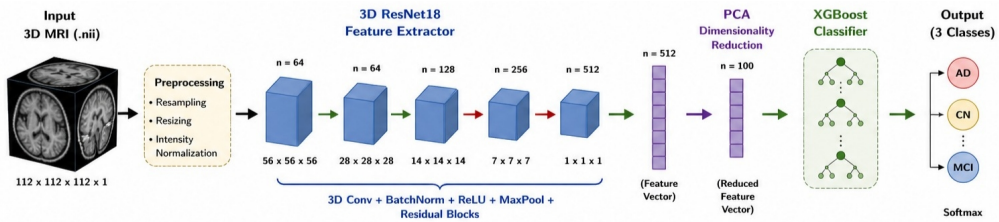


Figure 2: XGBoost Classifier Architecture

3.2.3 Pretrained 3D ResNet18

In the third model, the pretrained 3D ResNet18 architecture was used in combination with MONAI. To overcome the vanishing gradient problem, ResNet incorporates residual learning. This architecture enables efficient training of deep networks by allowing identity mappings. Transfer learning involves using pre-trained weights from a vast dataset, allowing the network to use existing knowledge. The network is fed with a 3D MRI volume and is employing multiple residual blocks to obtain hierarchical characteristics. The characteristics are next transmitted into a Global Average Pooling layer, with a fully connected layer for classification. The outcome is a

softmax activation function that gives the likelihood of a class. This model was trained with Adam optimizer with learning rate 0.0001, batch size 4 and 30 epochs.

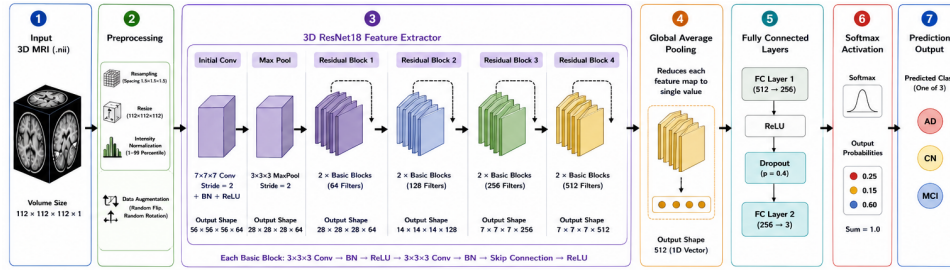


Figure 3: 3D ResNet18 Model Architecture

3.2.4 Radiomics-based Machine Learning Model

The fourth approach utilizes a radiomics-based machine learning methodology which involves handcrafted feature extraction methods for the classification problem of Alzheimer’s disease. MRI data was used for extracting statistical, texture-based and Gray-Level Co-occurrence Matrix (GLCM) radiomics features to gain fine detail of the tissue structure and heterogeneity. To tackle the dimensionality of features, dimensionality reduction techniques such as Principal Component Analysis (PCA) were used to eradicate duplicate features and enhance the classification results. To classify Alzheimer’s disease as a multi-class problem, we used reduced feature set to train a LightGBM classifier. The model was trained using 800 iterations of boosting with an optimized learning rate of 0.03 for classification.

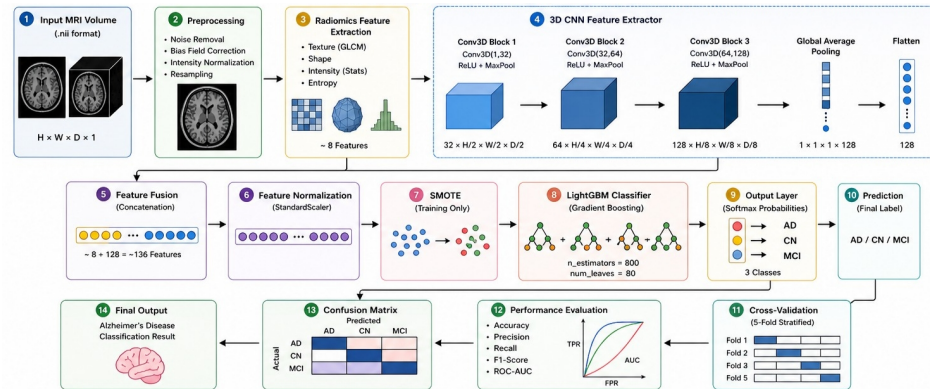


Figure 4: Radiomics-based Machine Learning Model Architecture

4 Experimental Results and Analysis

This section reports experiments and comparative studies of the proposed models for the classification of multiple classes of AD using 3D MRI data. The model’s performance was monitored utilizing Accuracy, Precision, Recall, and F1-score. Four different approaches were analyzed, including a 3D DenseNet with self-attention, 3D ResNet18, XGBoost classifier, and a radiomics-based machine learning model. The experiments were conducted to assess the efficacy of volumetric deep learning and handcrafted feature-based methodologies in distinguishing between Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer’s Disease (AD) classes.

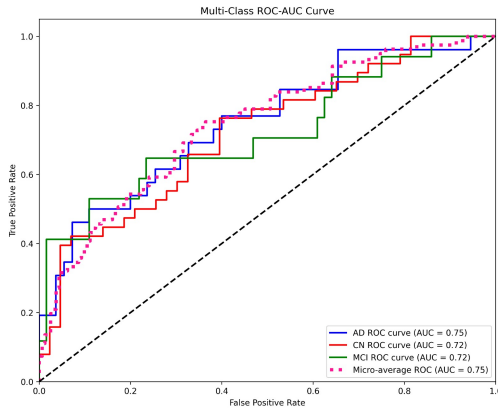
Table 1 presents the comparative performance evaluation of all implemented models. Among the evaluated approaches, the pretrained 3D ResNet18 achieved the highest overall performance with an accuracy of 0.61, recall of 0.61, precision of 0.60, and F1-score of 0.60. The findings show that transfer learning with pre-trained 3D architectures helps to optimize retrieval of attributes and generalization ability for classification of

Alzheimer’s disease from volumetric MRI imaging. The 3D DenseNet with self-attention and XGBoost classifier achieved comparable accuracy values of 0.57. The self attention mechanism and enhanced feature reuse in the 3D DenseNet provided a slight improvement in the balanced performance across the evaluation metrics. The radiomics-based machine learning model outperforms neither other models nor human experts, and it had an accuracy of 0.53, recall of 0.49, precision of 0.50, and F1-score of 0.50, implying that it is not able to capture the complex spatial representation as effectively as the deep learning-based models.

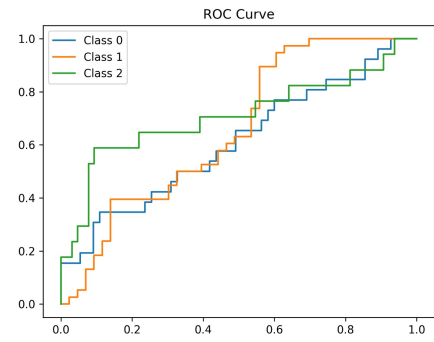
Models	Accuracy	Recall	Precision	F1 Score
3D DenseNet With Self-Attention Model	0.57	0.57	0.56	0.57
XGBoost Classifier	0.57	0.52	0.58	0.52
Pretrained 3D ResNet18	0.61	0.61	0.60	0.60
Radiomics based Machine Learning	0.53	0.49	0.50	0.50

Table 1: Comparative performance analysis of implemented models.

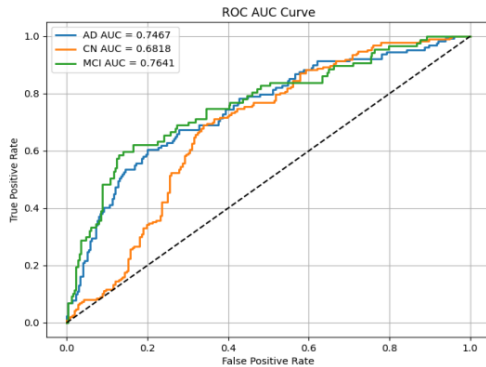
For all implemented models, ROC analysis was used to assess the discriminative ability for the multi-class classification of Alzheimer’s disease. The ROC analysis shows the efficiency of the model in categorizing various phases of Alzheimer’s disease based on volumetric MRI data.



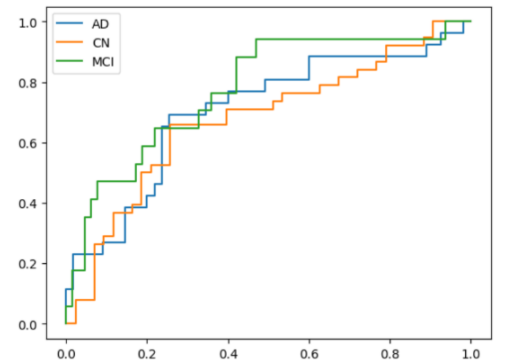
(a) 3D DenseNet with Self-Attention



(b) XGBoost Classifier



(c) Pretrained 3D ResNet18



(d) Radiomics-Based Model

Figure 5: ROC Curve of the implemented classification models.

The confusion matrix for the 3D ResNet18 model also shows the classification accuracy for each class. The model achieved better classification performance for the CN class while demonstrating improved detection capability for AD cases compared to the other implemented models. However, some misclassification between MCI and AD samples was still observed, highlighting the difficulty of distinguishing closely related disease stages in multi-class Alzheimer’s disease classification tasks.

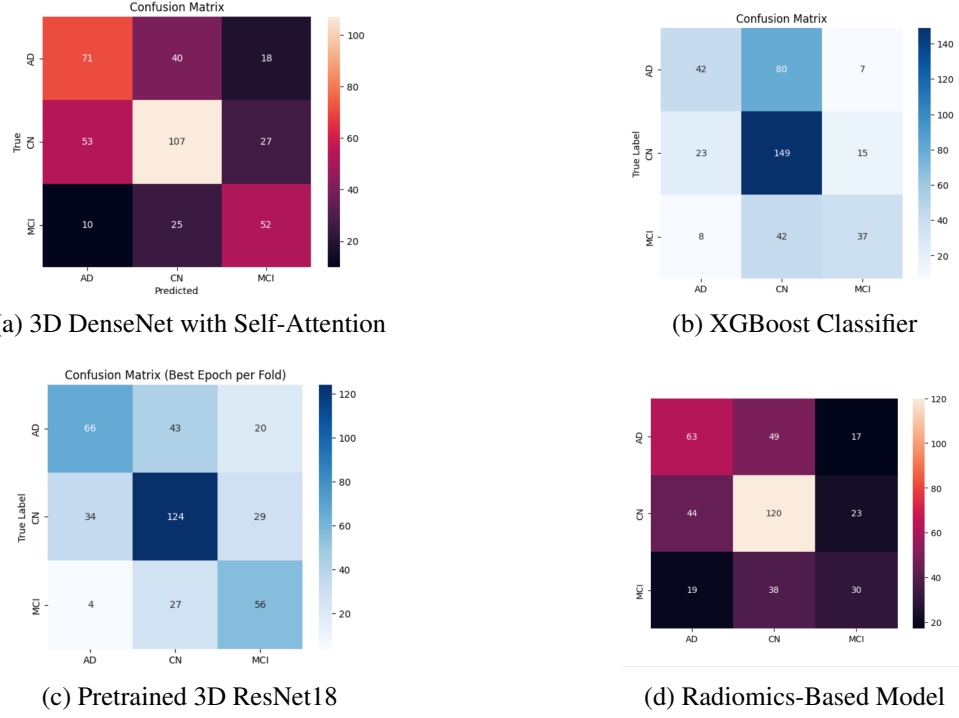


Figure 6: Confusion Matrix of the implemented classification models.

5 Conclusion

In this study, we evaluated the effectiveness of four different models for multi-class classification of Alzheimer’s disease stages using 3D MRI data, including a 3D DenseNet with self-attention, a pretrained 3D ResNet18, XGBoost classifier, and a radiomics-based XGBoost model. The experimental results showed that the accuracy, recall, precision, and F1-score were 0.61, 0.61, 0.60, and 0.60 respectively for the pretrained 3D ResNet18 among all the evaluated models. The 3D DenseNet model with self-attention and XGBoost classifier had an accuracy of 0.57, whereas the machine learning model using radiomics had an accuracy of 0.53. The results show that transfer learning based 3D deep learning architectures outperform the other architectures in terms of feature representation and classification accuracy at Alzheimer’s disease detection tasks. In comparison, it also demonstrated that deep learning models were able to more accurately extract complex spatial patterns from volumetric MRI than handcrafted radiomics features and traditional machine learning methods. The pretrained 3D ResNet18 achieved more balanced classification performance over all the three classes of CN, MCI and AD; whereas, the radiomics-based and standalone XGBoost methods exhibited relatively poor performance in classification of the similar disease stages.

Despite the promising results, several limitations were observed during this study. A key constraint was the relatively small number of images available for each of the disease classes, which made it difficult for the models to learn well in all classes. Different acquisition protocols, type of MRI scanner, and image quality might have also influenced classification accuracy. Moreover, using the 3D volumetric data substantially expanded both the computational load and the memory usage and training time, making it harder to run experiments with larger architectures and hyperparameter optimization. Distinguishing between very similar stages like MCI and AD was another challenge, as they share overlapping patterns of structural brain changes.

Future work may involve maximizing the robustness and generalization of models using larger and more varied datasets captured from different imaging centers. A combination of predictions from various classifiers using ensemble learning methods can achieve better classification accuracy and stability. Another approach to dealing with the lack of labeled medical data is the use of advanced data augmentation and synthetic data generation methods like Generative Adversarial Networks (GANs). Additionally, embedding Explainable

Artificial Intelligence (XAI) techniques such as attention maps and saliency visualization into the model could enhance interpretability and support medical decisions.

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